# What Factors Drive High IMDB Ratings and Box Office Revenue?

Phoebe Zhang Tushar Dhanorkar Isha Mantri

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## 1 Executive Summary

This report analyses the IMDB Top 1000 Movies dataset to identify the key factors driving high audience ratings and box office revenue, revealing that film success is fundamentally two-dimensional. Genre emerges as the strongest differentiator, Adventure and Sci-Fi films dominate commercially while Western, War, and Film-Noir genres earn the highest audience ratings confirming a clear divide between what makes money and what wins critical appreciation. Runtime, director and cast also play meaningful roles, with longer films and directors like Christopher Nolan linked to stronger ratings, while franchise driven directors like Anthony Russo

and actors like Robert Downey Jr. lead in box office revenue. These findings suggest that studios chasing critical recognition should invest in narrative-driven genres with experienced directors, while those prioritising commercial returns should continue developing large-scale franchise productions, and that the industry as a whole would benefit from adopting a dual-metric approach that values both IMDB ratings and box office revenue equally.

## 2 Introduction

Before COVID-19 turned everything upside down, the global film industry was pulling in around $26 billion USD at the box office every year. IMDB the go-to site for movie ratings gets over 250 million visitors a month, with more than 1.3 billion users worldwide sharing their views on films. So what actually makes a movie successful? It is a harder question than it sounds some films get rave reviews but barely sell a ticket, while others break box office records without a single award to their name. We look at success from two sides: how good people think a film is, through IMDB ratings and Metascores, and how much money it actually makes at the box office. Things like genre, runtime, director, cast and the decade a film was released in are all thought to play a role. To find out which ones actually matter, we used the IMDB Top 1000 Movies dataset from Kaggle (Shankhdhar (2021)) 1000 of the highest-rated films spanning from the 1920s all the way to the 2020s. All the analysis and visualisation was done in R (R Core Team (2024)). Our aim is simple figure out what really drives film success and give filmmakers and studios something practical to work with.

## 3 Data

The dataset used in this analysis is the IMDB Top 1000 Movies dataset, sourced from Kaggle (Shankhdhar (2021)). It contains information on the 1000 highest-rated films on IMDB, spanning nearly a century of cinema from the 1920s to the 2020s

The key variables used in this analysis include:

- **Runtime** - how long the film runs, measured in minutes
- **Genre** - the main category the film falls under, such as Action, Drama or Comedy
- **IMDB Rating** - the average score given by users on IMDB, rated on a scale of 1 to 10
- **Metascore** - the critic rating from Metacritic, scored out of 100
- **Number of Votes** - the total number of users who rated the film on IMDB
- **Gross Revenue** - the total worldwide box office earnings in USD
- **Director** - the person who directed the film

- **Stars** - the lead actors featured in the film
- **Release Year** - the year the film was officially released

A few cleaning decisions were made before the analysis began. Films that had no gross revenue recorded were left out of the commercial performance comparisons - keeping those in would have skewed the results. We also noticed that some films had multiple directors listed on IMDB, so we kept only the first listed director for consistency. It's worth noting though that this created a small issue - in some cases, the additional directors ended up being incorrectly recorded as actors in the dataset, which is something to keep in mind when reading the director and actor findings.

# 4 Methodology

## 4.1 Data Preparation

Before jumping into the analysis, we did some tidying up of the data. Any film that was missing a gross revenue figure was removed from the commercial performance side of things - keeping them in would have made the averages unreliable.

We also ran into a quirk with the director data. Some films on IMDB have more than one director listed, but since the dataset wasn't structured consistently to handle that, we kept only the first director mentioned for each film. The tricky part is that the extra directors didn't just disappear - they ended up being misclassified as actors in the dataset instead. This is worth keeping in mind when looking at the director and actor results, as it may have affected how accurate those findings are.

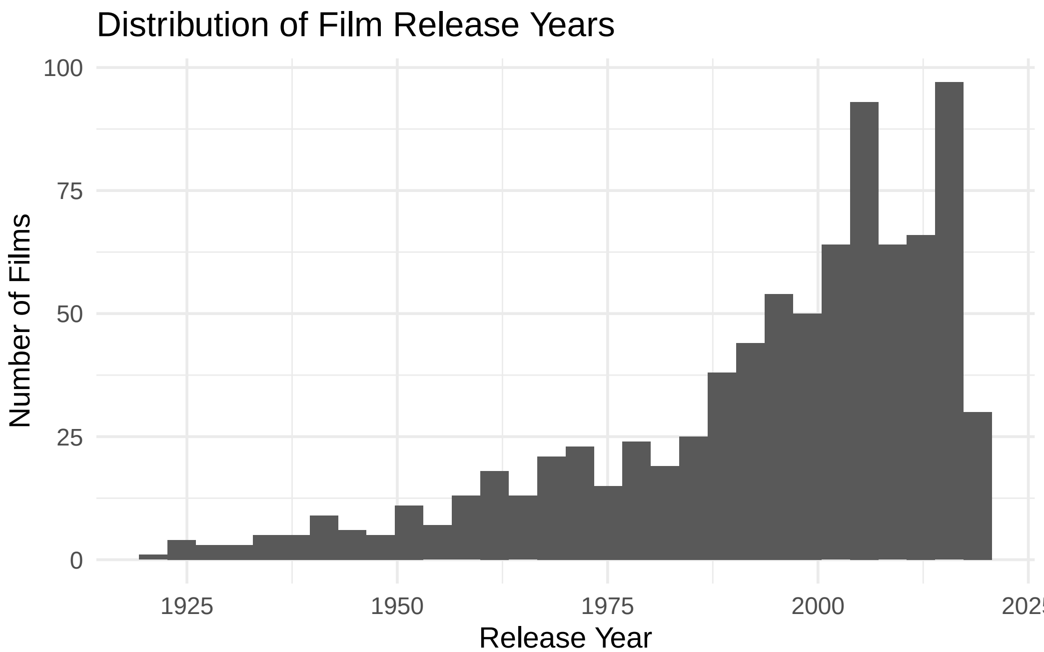


Figure 1: Distribution of Release Years in the Dataset

The temporal distribution shown in (Figure 1) displays a strong skew toward more recent films, meaning older films are underrepresented, limiting the ability to generalise conclusions across different film eras.

Runtime was grouped into four categories to allow comparison across film lengths:

- Less than 90 minutes
- 90 to 120 minutes
- 120 to 150 minutes
- More than 150 minutes

## 4.2 Analysis Approach

The analysis examines two measures of film success separately:

- **Perceived quality** -measured using average IMDB user ratings
- **Commercial performance** -measured using average gross box office revenue in USD

The following factors were examined against both measures:

- `Genre` -Average gross revenue compared across genres (Figure 1) and average IMDB rating compared across genres (Table 1)
- `Runtime` - Average IMDB rating (Figure 2) and average gross revenue (Figure 3) compared across runtime groups
- `Metascore` - Relationship between critic score and IMDB rating examined using a scatter plot (Figure 4)
- `Number of votes` - Relationship between vote count and IMDB rating examined using a scatter plot (Figure 5)
- `Director` - Top directors ranked by average IMDB rating (Figure 6) and average gross revenue (Figure 7)
- `Actor` - Top actors ranked by average IMDB rating (Table 2) and average gross revenue (Figure 8)
- `Release year` - Distribution of films across release years examined to identify temporal patterns in the dataset (Figure 9)

All figures and tables are produced using tidyverse (ggplot2) (Wickham et al. (2019)) and knitr (Xie (2024)) within a reproducible Quarto document (Posit Software (2024)).

# 5 Results

The analysis of the Top 1000 IMDb rated films highlights clear differences between films that performed well critically and those that performed well commercially.

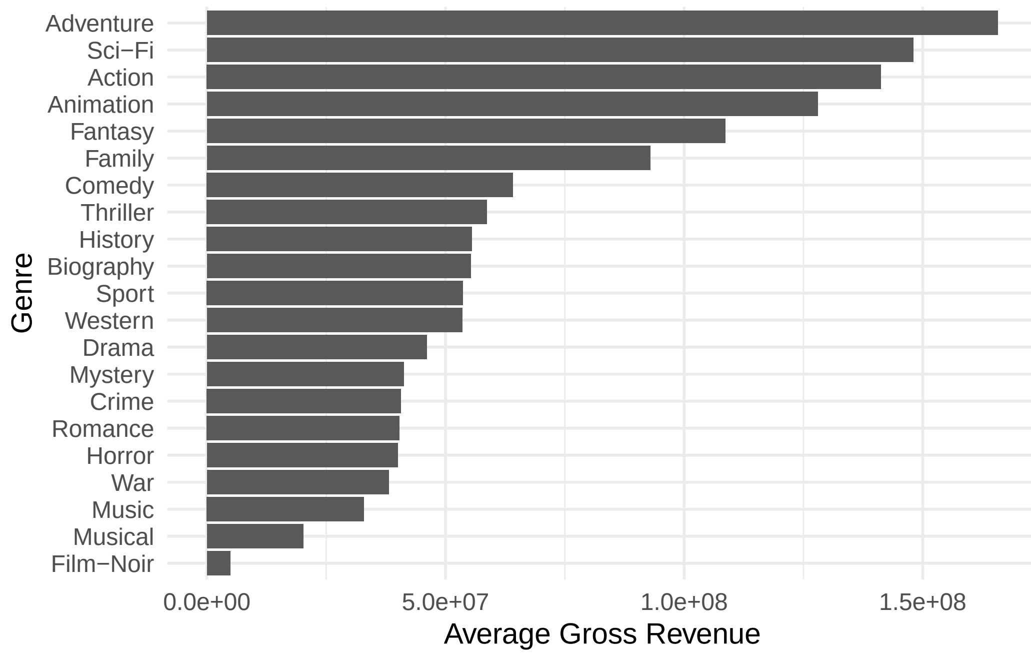


Figure 2: Top average grossing genres

As shown in Figure 2, Adventure and Sci-Fi genres record the highest average gross revenue.

Table 1

| Genre | Average Rating |
|---|---|
| Western | 8.04 |
| War | 8.02 |
| Film-Noir | 8.01 |
| Sci-Fi | 7.99 |
| Mystery | 7.97 |
| Adventure | 7.96 |
| Crime | 7.96 |
| Drama | 7.95 |
| Action | 7.95 |
| Romance | 7.93 |

In contrast, Western and War films achieve the highest average IMDb ratings, as shown in Table 1.

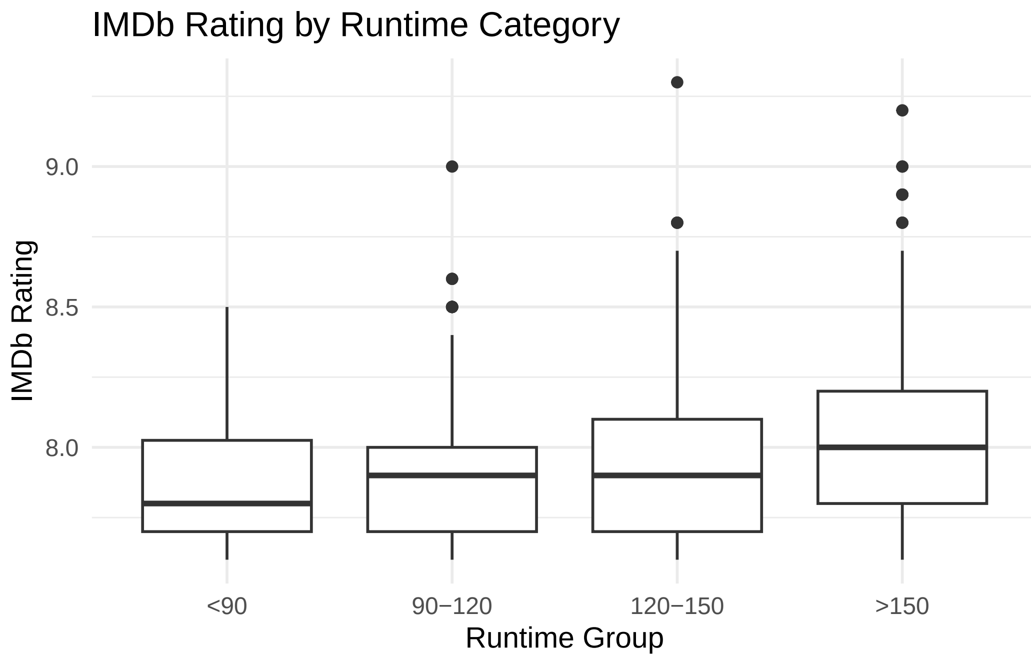


Figure 3: Average Ratings per Runtime Group

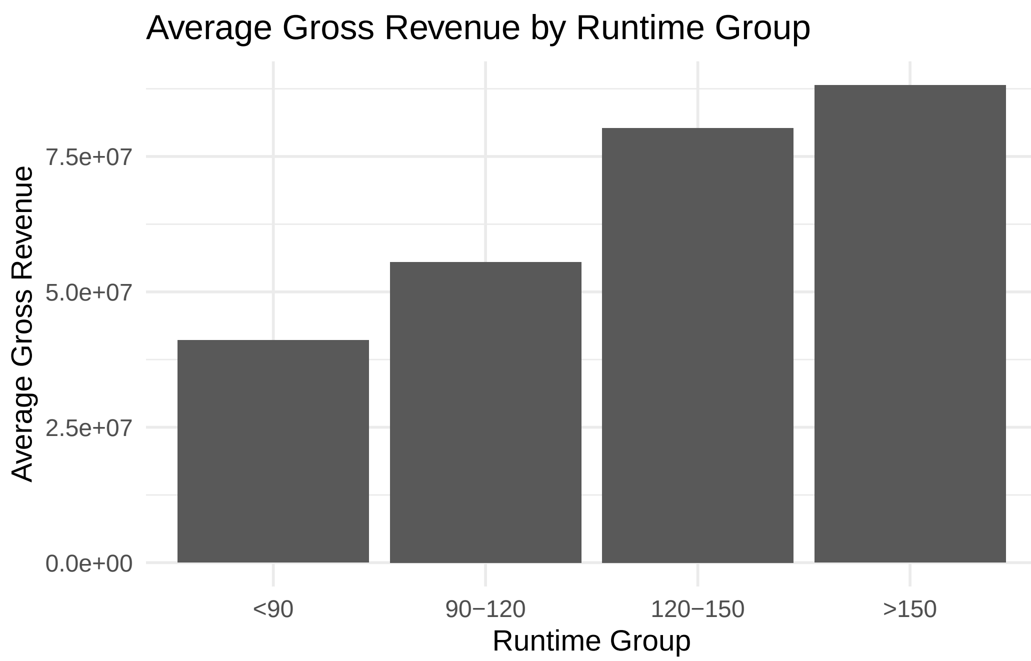


Figure 4: Average Gross Revenue per Runtime Group

Films with longer runtimes (above approximately 150 minutes) tend to have slightly higher average IMDb ratings (Figure 3) and perform better in the box office (Figure 4).

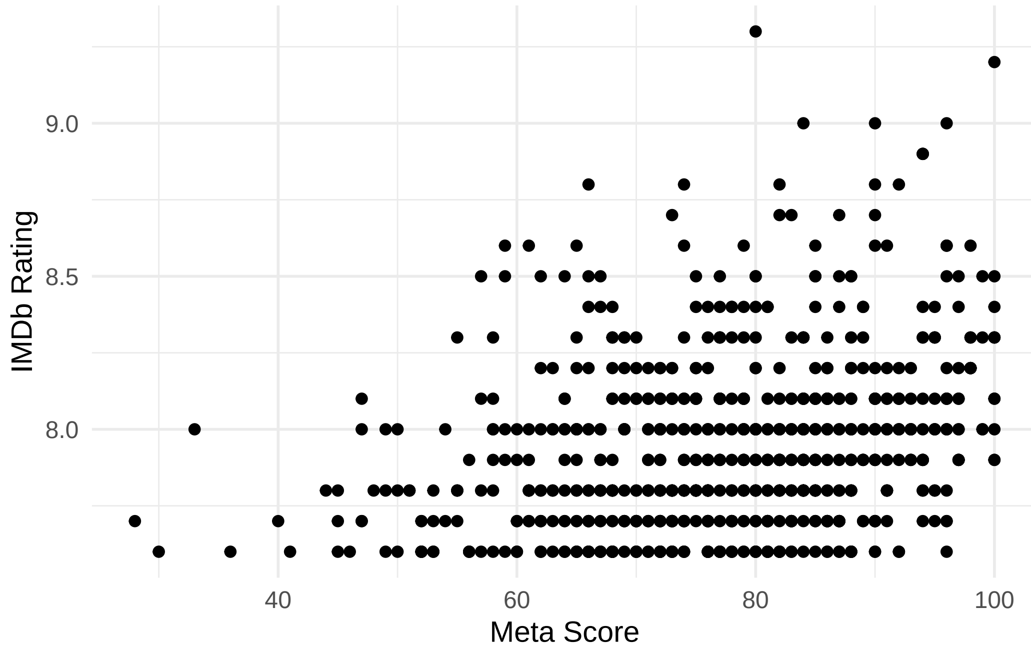


Figure 5: Relationship between Meta score and IMDb Rating

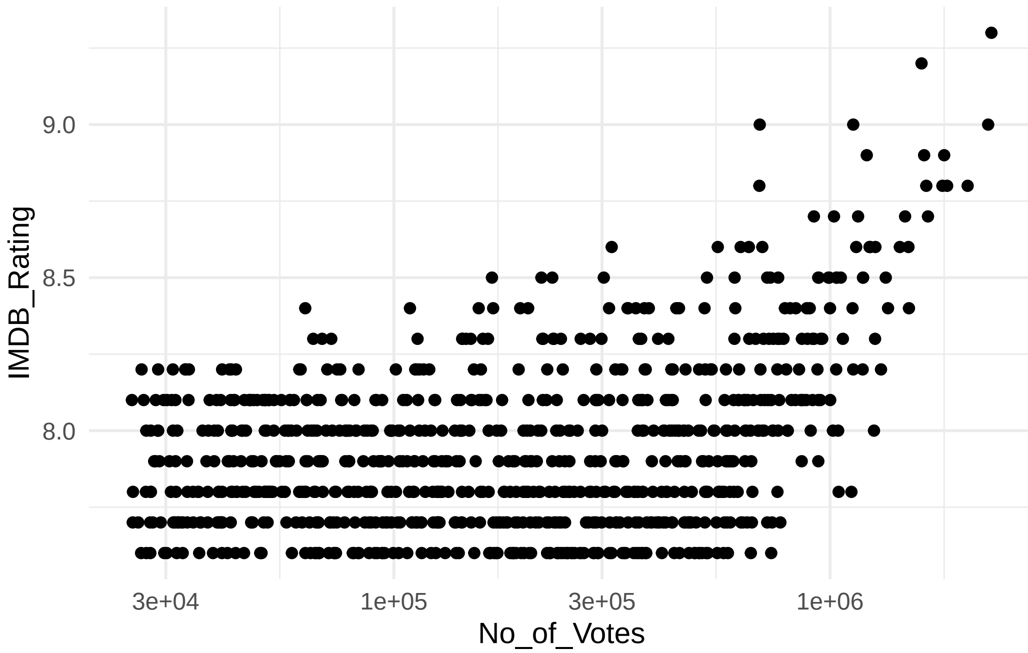


Figure 6: Relationship between Number of Votes and IMDb Rating

A positive relationship is observed between Meta score and IMDb rating (Figure 5). Similarly, the number of votes is positively associated with IMDb ratings (Figure 6).

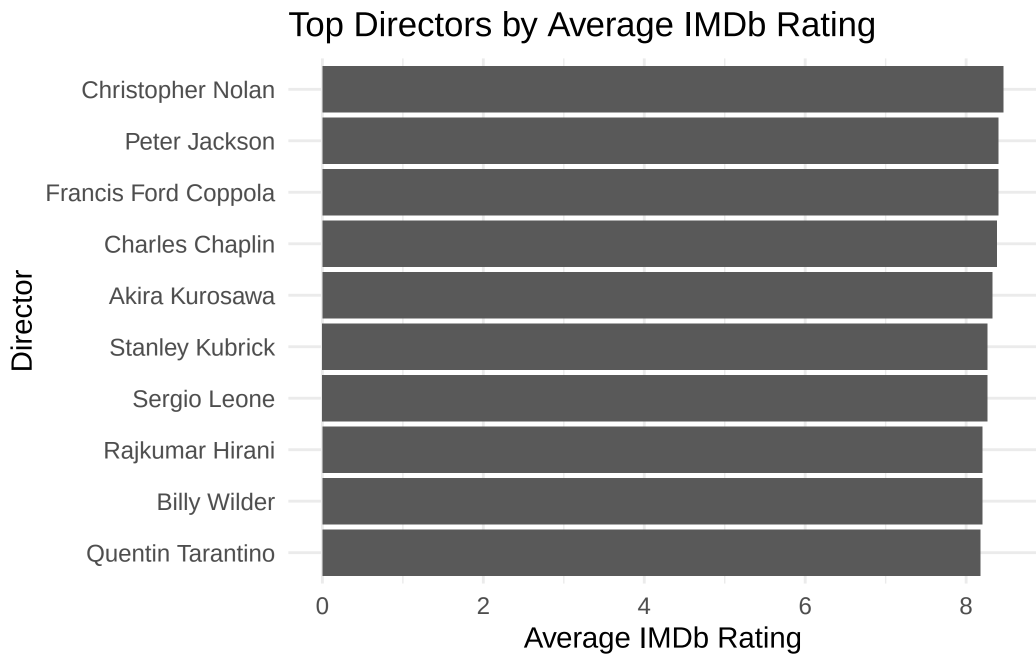


Figure 7: Directors with Highest Average IMDb Rating

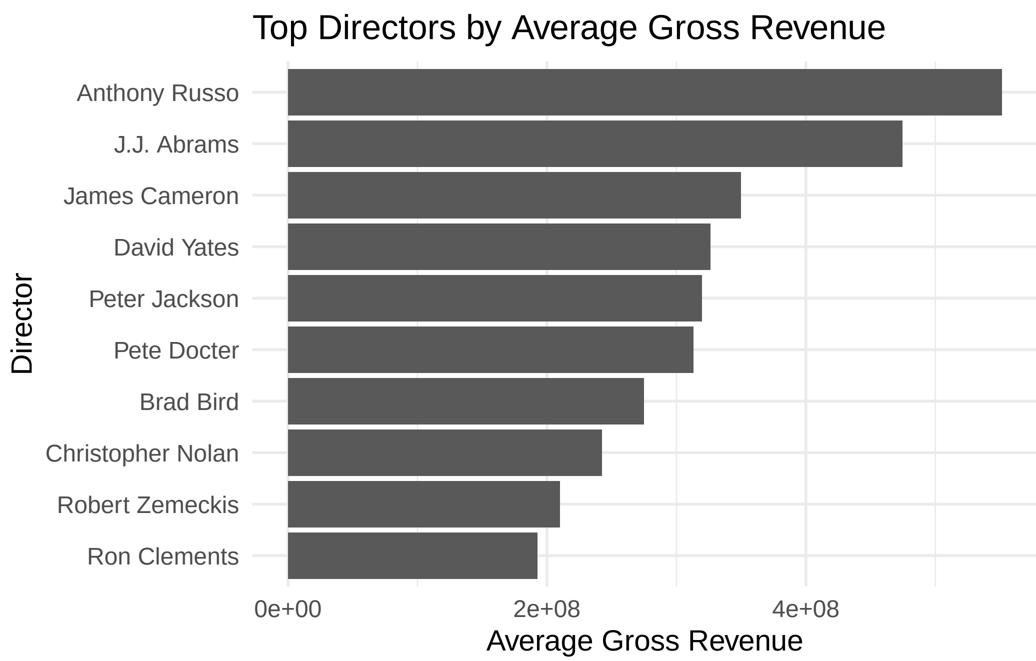


Figure 8: Directors with Highest Average Gross Revenue

Director-level results show that Christopher Nolan films have the highest average IMDb rating 8.46 (8.46) (Figure 7), while Anthony Russo films have the highest average gross revenue (Figure 8).

Table 2: Actors with Highest Average IMDb Rating

| Actor | Number of Movies | Average Rating |
|---|---|---|
| Elijah Wood | 3 | 8.80 |
| Orlando Bloom | 4 | 8.60 |
| Mark Hamill | 3 | 8.53 |
| Marlon Brando | 4 | 8.43 |
| Charles Chaplin | 5 | 8.38 |
| Lee J. Cobb | 3 | 8.37 |
| Takashi Shimura | 3 | 8.37 |
| Diane Keaton | 5 | 8.34 |
| Alec Guinness | 3 | 8.33 |
| Robert Duvall | 3 | 8.33 |

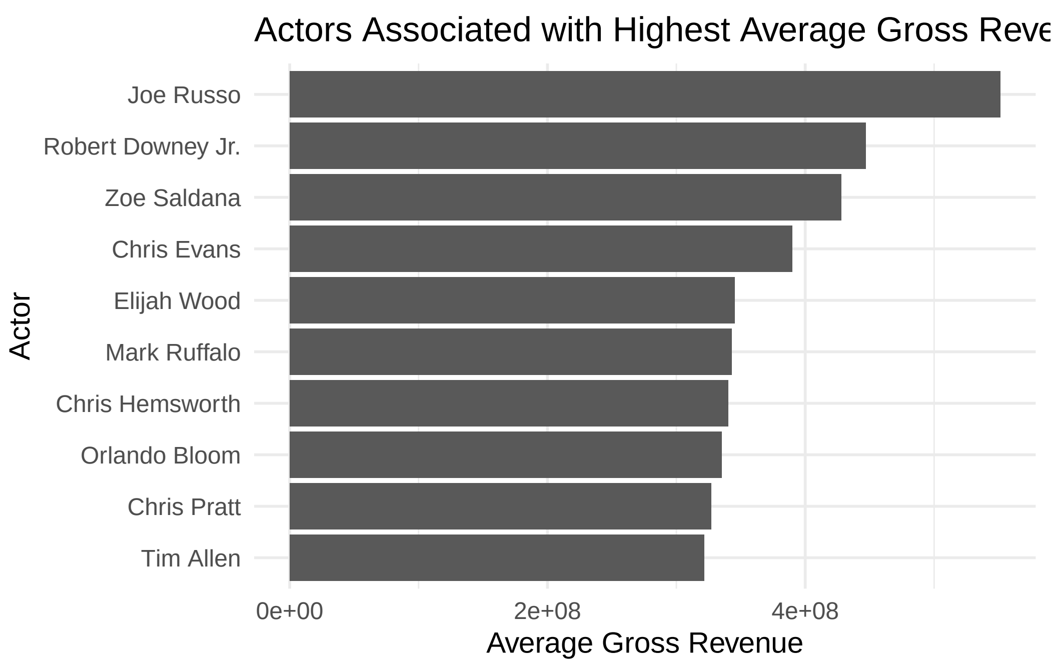


Figure 9: Actors with Highest Average Gross Revenue

Actor-level results indicate that Elijah Wood and Orlando Bloom appear in films with higher average IMDb ratings (Table 2), while Robert Downey Jr. and Zoe Saldana are associated with higher average gross revenue (Figure 9).

# 6 Discussion

The results suggest that film success is multi-dimensional, with different factors contributing to commercial performance and perceived quality. The moderate positive relationship between Meta score and IMDb rating indicates partial alignment between critic evaluations and audience perceptions. However, substantial variation suggests that audiences do not always agree with critics in their assessments of film quality.

The analysis of the number of votes further supports this distinction. Films with higher vote counts tend to have higher IMDb ratings, although this relationship is likely driven by audience exposure rather than intrinsic quality. Films that are more widely viewed accumulate more ratings, which increases the stability and reliability of the observed IMDb scores.

Adventure and Sci-Fi films tend to generate the highest average gross revenue, reflecting the commercial appeal of visually driven and large-scale productions. However, higher revenue does not necessarily correspond to higher IMDb ratings, reinforcing the divergence between commercial success and perceived quality.

Runtime shows that longer films tend to receive slightly higher IMDb ratings. This may reflect the potential for more complex storytelling and character development in longer films, which audiences may value more.

At the director level, the results suggest differences in performance across individuals, with Christopher Nolan associated with higher average IMDb ratings and Anthony Russo associated with higher average gross revenue. This may reflect differences between critically acclaimed filmmaking and franchise-based commercial success.

Similarly, actor-level analysis suggests variation in associations with both ratings and revenue. Some actors are more strongly associated with high-grossing films, while others appear more frequently in highly rated films. This pattern is consistent with the concept of star power, where well-known actors contribute to commercial visibility and box office performance. However, these findings should be interpreted cautiously due to uneven representation across films and potential dataset inconsistencies in the underlying data structure.

A key limitation of this study is that the dataset is restricted to the IMDb Top 1000 films, which introduces selection bias towards already successful or highly rated titles. As a result, the findings are not representative of the broader population of all films.

Furthermore, in the original scraping process, the reliability of the director and star level data was not consistently structured. During the cleaning process, we realised that for films where multiple directors were listed on IMDb, only the first listed director was retained, while additional directors were incorrectly categorised as actors. This reduces the precision of the conclusions drawn from the director and actor analysis and thus the interpretations in the results section may not truly reflect the displayed figure.

Overall, the results indicate a clear distinction between audience evaluation (IMDb rating) and commercial success (gross revenue), with different factors influencing each outcome. These patterns are consistent with the idea that commercial success is more closely related to visibility and celebrity influence, while IMDb ratings reflect audience assessment of the film aspects itself.

## 7 Conclusion

This report examined the factors associated with film success using a dataset of the IMDb Top 1000 films, with success defined in terms of both commercial performance (gross revenue) and perceived quality (IMDb ratings). The findings demonstrate that film success is multi-dimensional, with clear differences in the determinants of critical reception and financial performance.

Overall, the results indicate only a moderate alignment between critic evaluations (Meta score) and audience perceptions (IMDb ratings), suggesting that critical and audience judgments are related but not equivalent. Similarly, while films with higher visibility tend to accumulate more votes and slightly higher ratings, this effect appears to reflect exposure rather than inherent quality differences.

Genre analysis highlights a strong commercial advantage for Adventure and Sci-Fi films, which achieve the highest average gross revenue. However, this does not translate into higher audience ratings, reinforcing a separation between commercial appeal and perceived quality. Runtime shows a relationship with both IMDb ratings and box office revenue, suggesting influence on both audience evaluation and commercial value.

At the individual level, differences between directors and actors further support the distinction between commercial and critical success. Some figures are more strongly associated with high-grossing films, while others are linked to higher-rated productions, reflecting different industry roles and production contexts such as franchise filmmaking versus director-led work.

However, these conclusions should be interpreted in light of important limitations. The dataset is restricted to top-rated films, introducing selection bias and limiting generalisability to the broader film industry. Additionally, the overrepresentation of recent films constrains historical comparison. Data quality issues in the scraping process, particularly the misclassification of multiple directors as actors, also reduce the precision of some role-based interpretations.

In summary, the analysis provides evidence that commercial success and perceived quality are influenced by different factors such that a film can be highly rated without earning the most revenue, and a commercially successful film may not always receive the highest ratings. These findings highlight the complexity of defining "success" in the film industry and demonstrate the importance of considering the combination of multiple factors when evaluating film performance.

# 8 Recommendation

**For Studios Prioritising Critical Acclaim**

Studios aiming for stronger critical and audience ratings should consider investing in Biography, Drama, and Animation films. These genres tend to perform well because they often focus on strong storytelling, emotional depth, and well-developed characters. Collaborating with experienced directors in these genres may also help improve the overall quality and reception of the film.

**For Studios Prioritising Box Office Revenue**

Studios focused on commercial success should continue to invest in Action, Adventure, and Science Fiction films. These genres often attract wider audiences and are well suited to large-scale production, franchise development, sequels, and strong marketing campaigns.

**For Independent Filmmakers**

Independent filmmakers should focus on genre-appropriate stories with clear narratives and strong creative direction. The findings suggest that success is not only dependent on budget or star power, but also on factors such as genre, runtime, and the ability to engage audiences through meaningful storytelling.

**For the Film Industry Broadly**

The industry should use a balanced approach when measuring film success. Instead of focusing only on box office revenue, studios and analysts should also consider IMDB ratings and Metascore. This would provide a fuller understanding of both short-term commercial performance and long-term audience value.

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